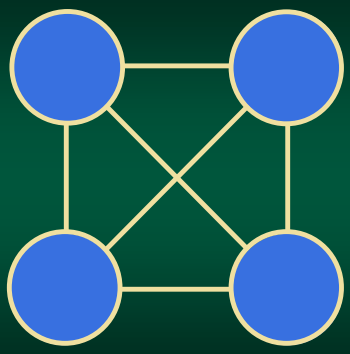
Graphs

Part 14

1



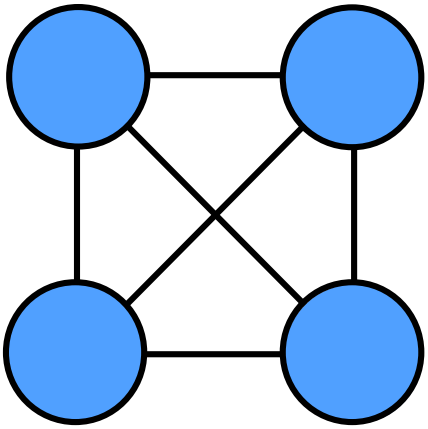
Introduction to Graphs

Nope, not the same as charts

2

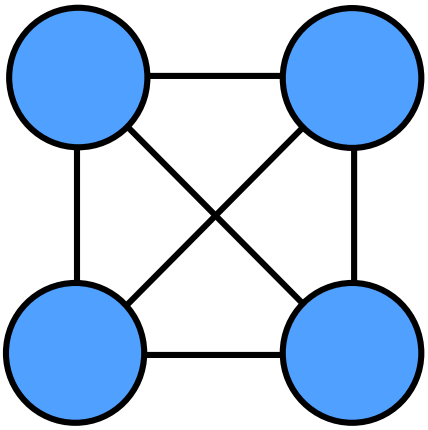
Graphs

- Lists and trees are just a special case of another structure - the *graph*
- Graphs are the basis for all of computer science



Graphs

- Computer science is not about chips, processors, etc...
- ... this is just implementation technology



Where are Graphs Used?

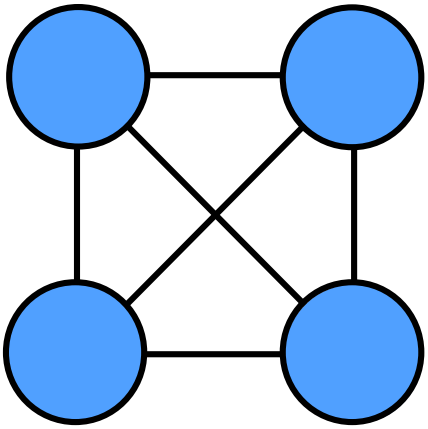
- The easy answer is: *everywhere*
- In computer science
  - state machines
  - mazes and networks
- Other fields
  - chemistry
  - physics
  - Government?

Motivation

- They are one of the pervasive data structures used in computer science
- Several real-life problems can be converted to problems on graphs
  - they are useful tool for modeling real-world problems
  - allows us to abstract details and focus on the problem

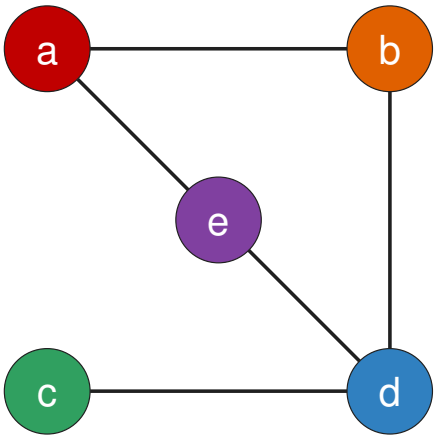
Terminology

- The terminology for graphs is a bit different from trees and linked lists
- Rather, it is more generalized
  - nodes are called *vertices*
  - branches are called *edges*



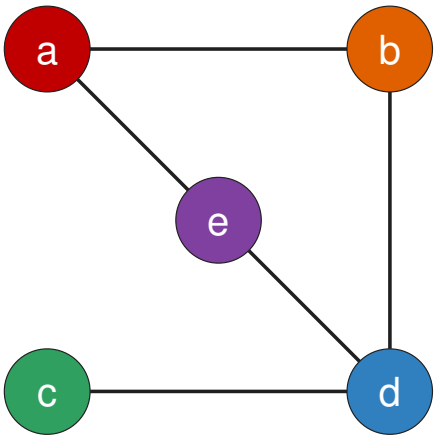
Example Graph – Vertices

- a
- b
- c
- d
- e



Example Graph – Edges

- a – b
- a – e
- b – d
- c – d
- d – e

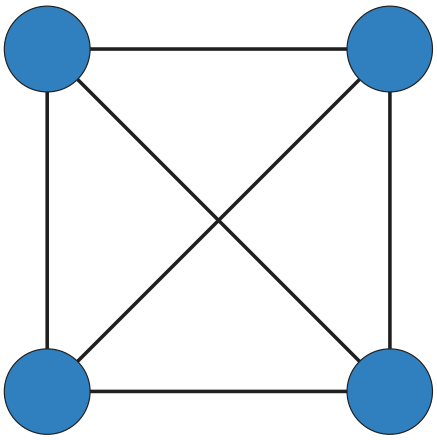


Adjacent and Incident

- When two vertices share an edge
  - they are said to be *adjacent*
  - in other words, they are connected
- The edge itself
  - is called *incident* on vertices x and y
  - in other words, it is the connection

Complete Graphs

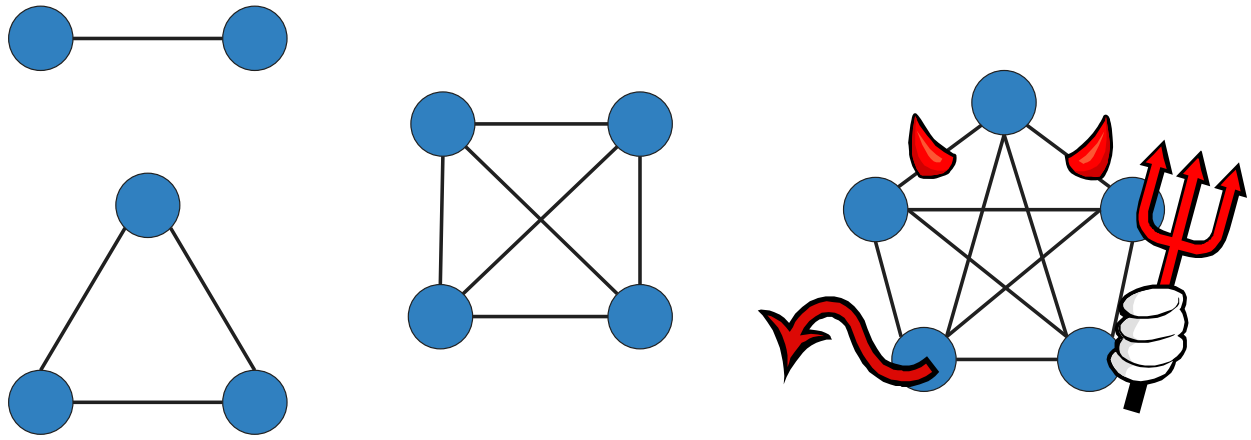
- A *complete graph* is one in which all pairs of vertices are adjacent
- In other words, every vertex is connected to every other vertex



Complete Graphs

- The # of edges in a complete graph...
  - if n is the total number of vertices, then each vertex is incident to $n - 1$ edges
  - we can compute $n \times (n - 1)$ edges, but this would count each edge twice!
  - so, the number of edges = $\boldsymbol{n \times (n - 1) / 2}$
- For a noncomplete graph...
  - number of edges < $n \times (n - 1) / 2$

Example Complete Graphs



Paths

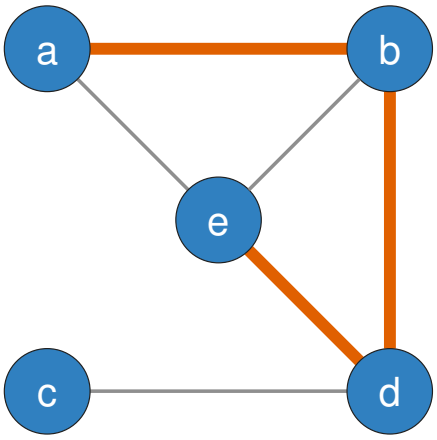
Some edges are one-way streets!

Paths

- A *path* is a sequence of vertices $v_1, v_2, \ldots v_k$ such that consecutive vertices v_n and v_{n+1} are adjacent
- This can represent
  - a physical path
  - logical connection
  - etc...

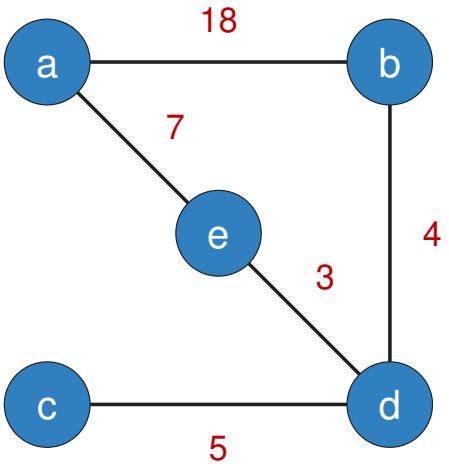
Paths

- In a *simple path*, all edges of a path are distinct
- The *length* of a path is measured by either the total number of edges or vertices



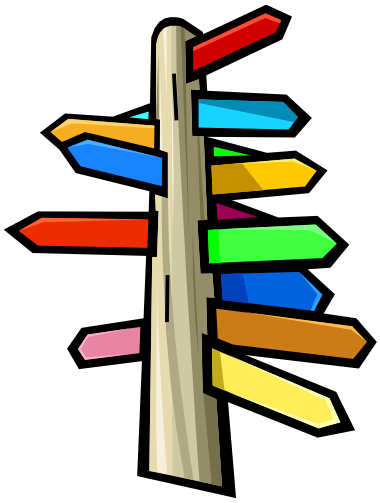
Weighted Graphs

- *Weighted graph* has values on each edge
- These values can be anything – and are defined by the ADT using the graph



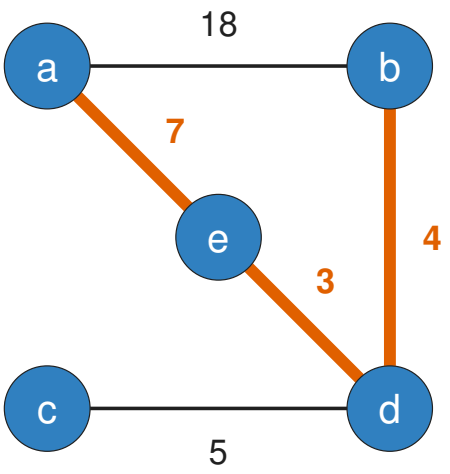
Weighted Graphs

- These weights are abstract and can represent *anything*
- Examples
  - distances – driving, flight paths
  - costs
  - server latency times
  - etc...



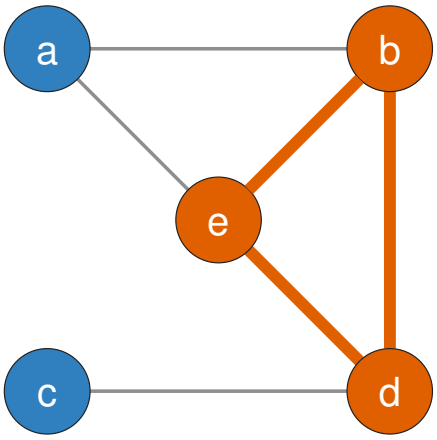
Minimum Path

- Often it is useful to find the minimum path – e.g. the smallest sum of edges
- Example: minimum path from a to b is:
  a → e → d → b
- We'll cover that soon!



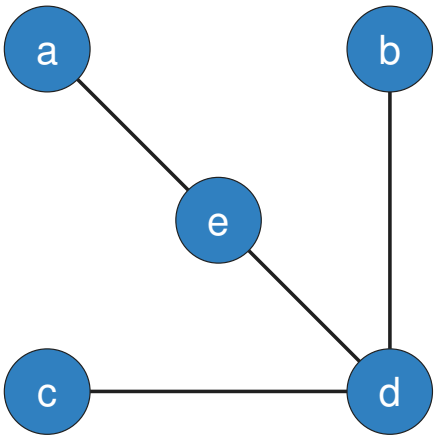
Cycles

- A *cycle* is sequence of vertices v_1, v_2, …v_k where…
  - v_n and v_{n+1} are adjacent
  - $v_1 = v_k$
- So, it's a loop of vertex



Cycles

- An *acyclic graph* does not contain any possible cycles
- This definition will come up later as an essential attribute for some solutions



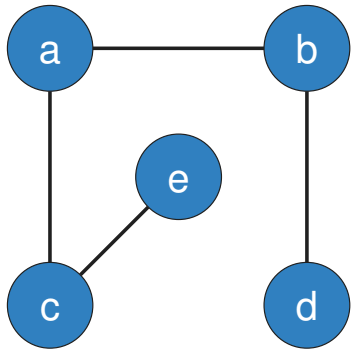
Connected and Unconnected

- A *connected* graph
  - has a path from every vertex to all other vertex
  - so, everything is connected somehow
- An *unconnected* graph
  - at least one vertex exists in which no path exists to another vertex
  - so, there are 2+ sub-graphs that are unlinked

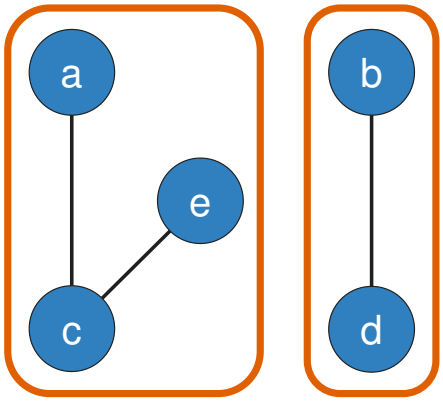
Connected and Unconnected

- The *connected component* is the maximum connected subgraph of a given graph
- If the graph is connected, then the whole graph is one single connected component

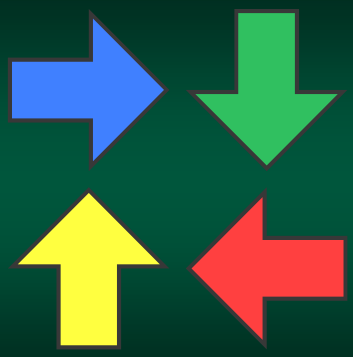
Connected and Unconnected



Connected



Unconnected



Directed & Undirected

Some edges are one-way streets!

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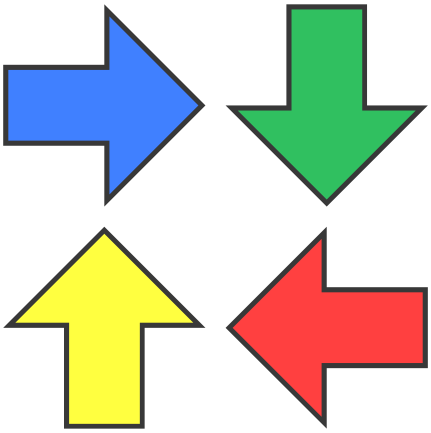
Directed & Undirected

- Sometimes, an edge can be travelled both directions
- However, sometimes, they have a distinct source and destination
- Think of it as streets – some are two-way, and some are one-way



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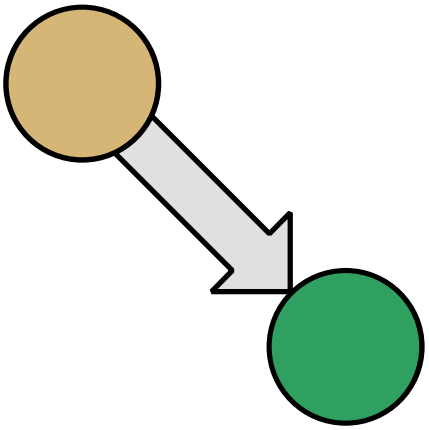
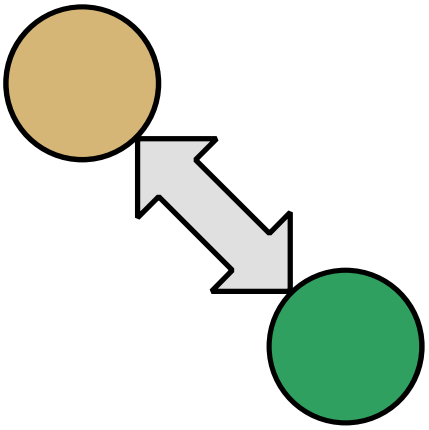
Directed Graph

- A *directed graph (digraph)* is a graph where each edge has a source and target vertex
- This is the basis most of the data structures used today:
  - trees
  - linked lists



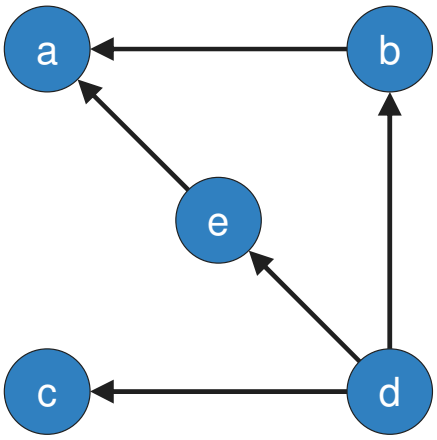
Undirected Graphs

- *Undirected graphs* have edges that have the <u>same</u> meaning for both directions
- So, if Vertex A links to Vertex B, then Vertex B also links to Vertex A

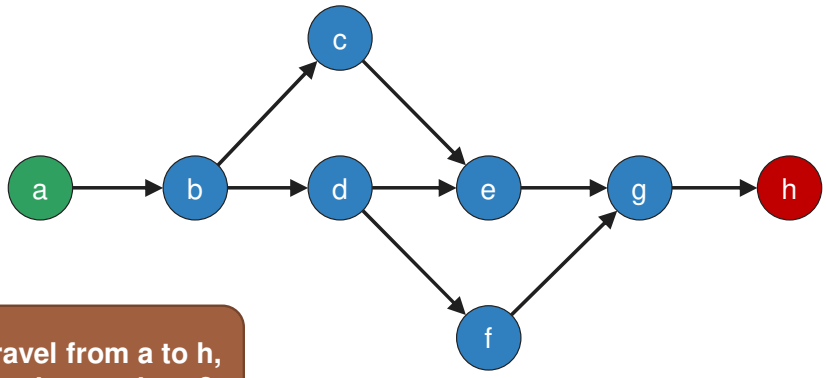


Directed Acyclic Graphs

- When a directed graph lacks cycles, it is called a *Directed Acyclic Graph* (DAG)
- Since edges only travel one way, this leads to some interesting graphs

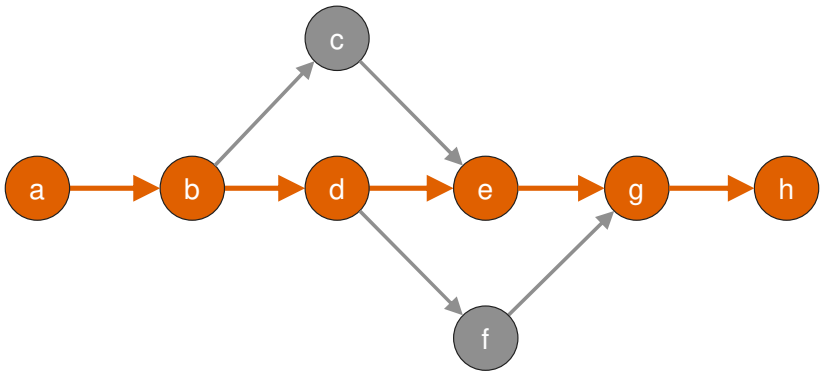


Examine this DAG: Path from a to h

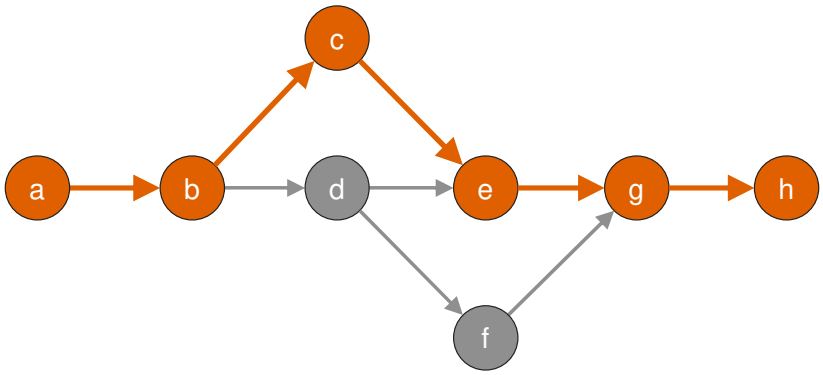


If we want travel from a to h, how many paths are there?

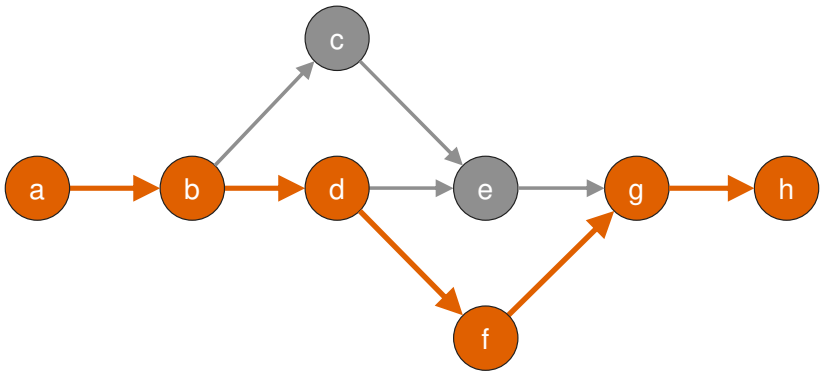
Examine this DAG: Path 1



Examine this DAG: Path 2

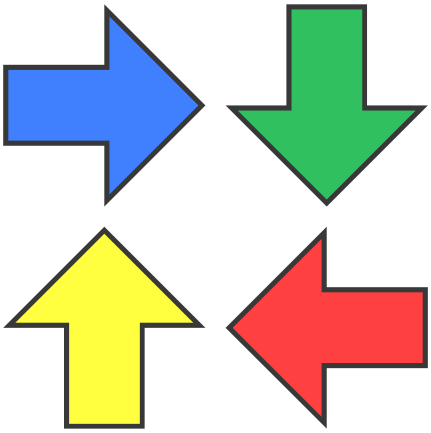


Examine this DAG: Path 3



Directed Acyclic Graphs

- In the last example, there are 3 paths from a to h
- Yet, there are **no** cycles
- So, while a DAG contains no cycles, there may be multiple paths from one vertex to another



Trees & Arborescence

A graph is a tree… if…

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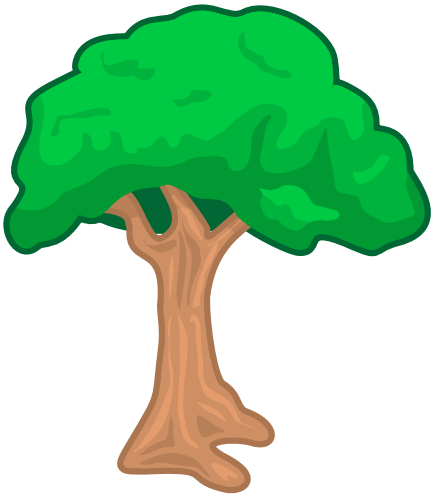
Arborescence

- An *arborescence* is an acyclic graph where, if there is a path from one vertex to another, it is **unique**
- So, there is at most one path from a vertex A to B



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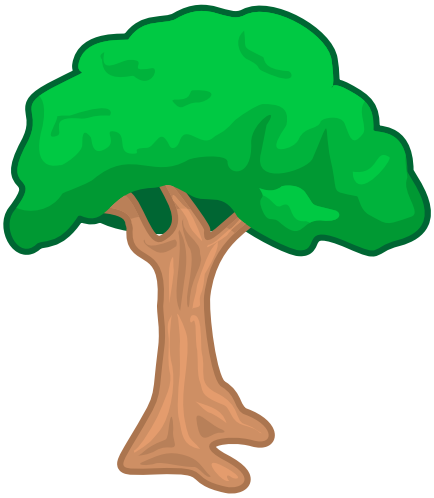
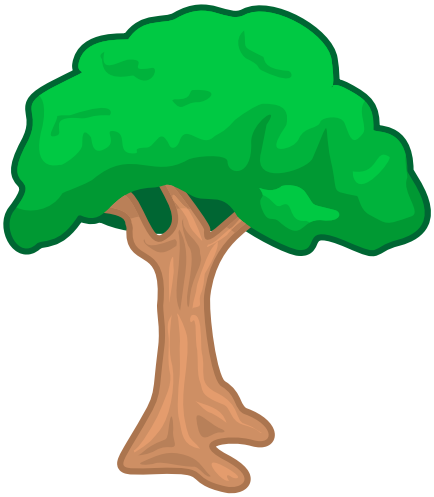
Alternative terms for Arborescence

- There are alternative terms (all of which are valid)
- Such as….
  - directed rooted tree
  - out-arborescence
  - out-tree
  - branching



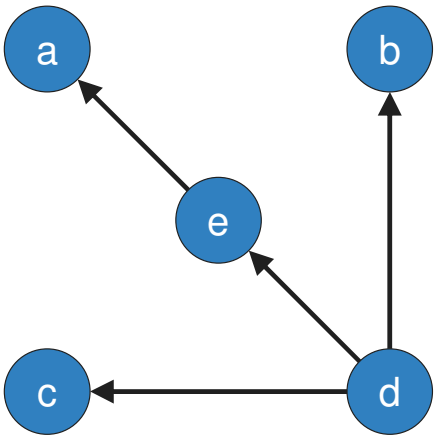
Trees

- All trees are acyclic graphs (which may be directed)
- But, not all DAGs are trees
  - DAGS allow multiple paths to the same vertice
  - since they directed, this won't create cycles

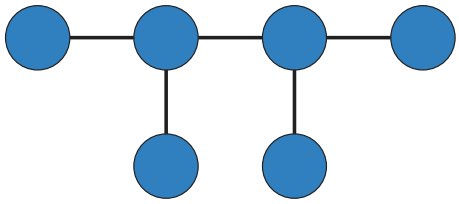


Trees

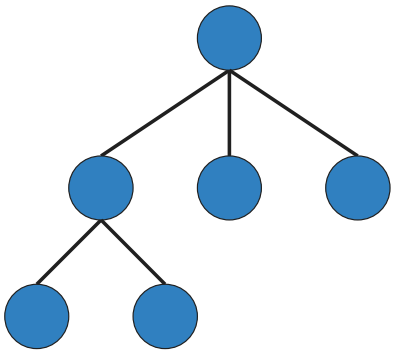
- *Rooted Tree* selects an arbitrary vertex as the root
- *Forest* is a collection of trees



Trees

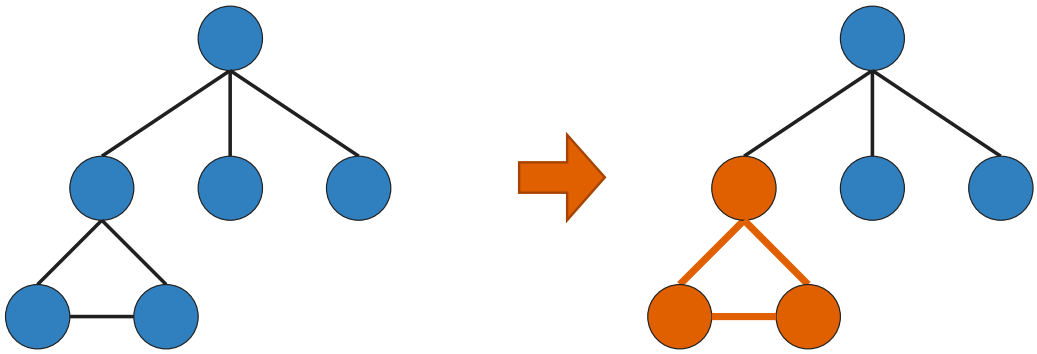


Tree

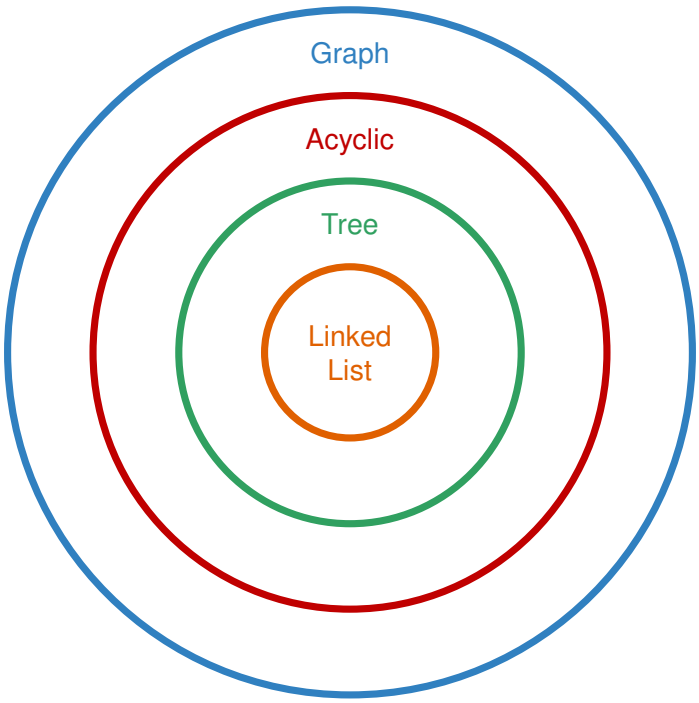


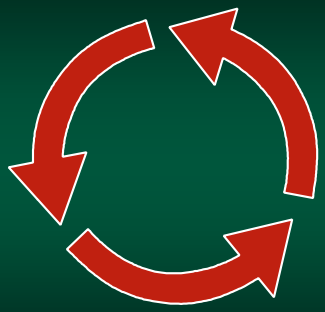
Same Tree

Trees



NOT a Tree





Graph Traversal

Not as easy as trees

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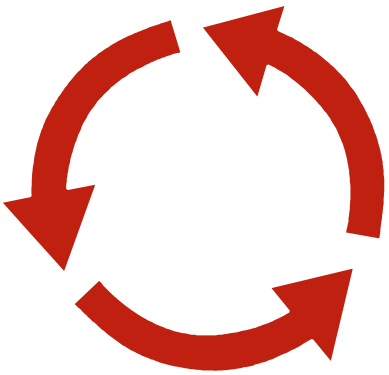
Graph Traversal

- Typically when you search a tree, you can use a simple depth-first search
- However, graphs can contain cycles
- Your program can get stuck in an graph loop and never escape
- This has to be taken into account



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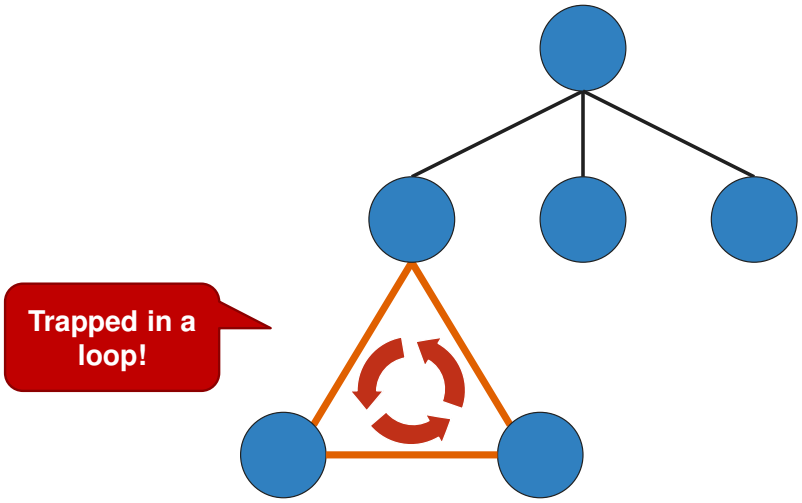
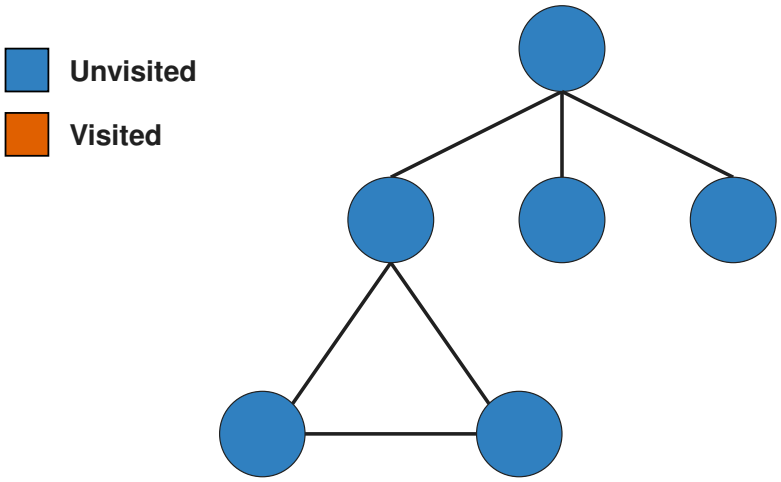
Graph Traversal



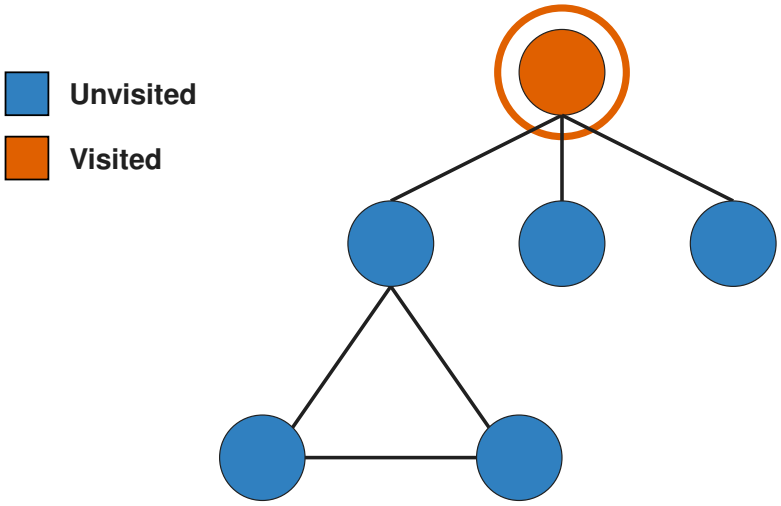
Graph Traversal

- So, to before you search…
  - the vertices need to visited only **ONCE**
  - otherwise, we have a loop
- *Solution:* vertex has a "visited" property
  - before the search, each vertex is set to false
  - when the search visits them, it is set true
  - search never follows an edge to a visited vertex

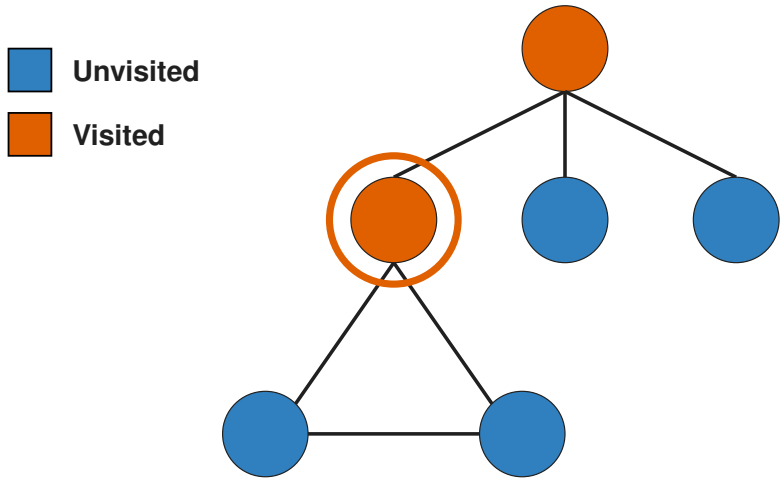
Graph Traversal



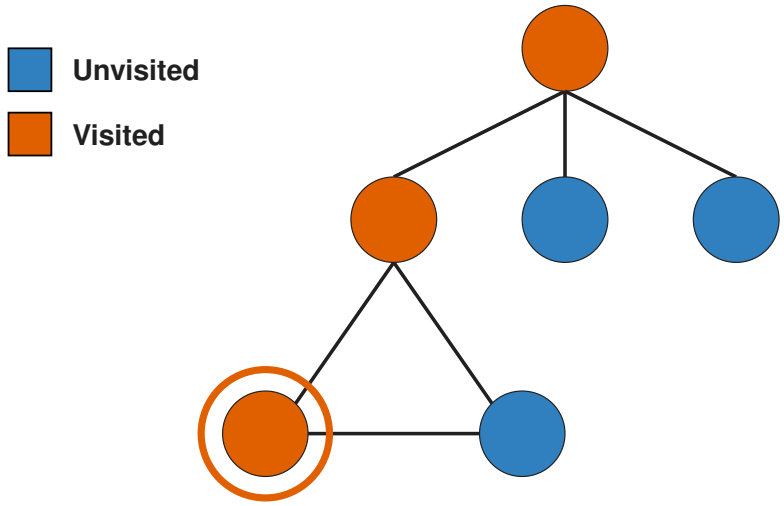
Graph Traversal



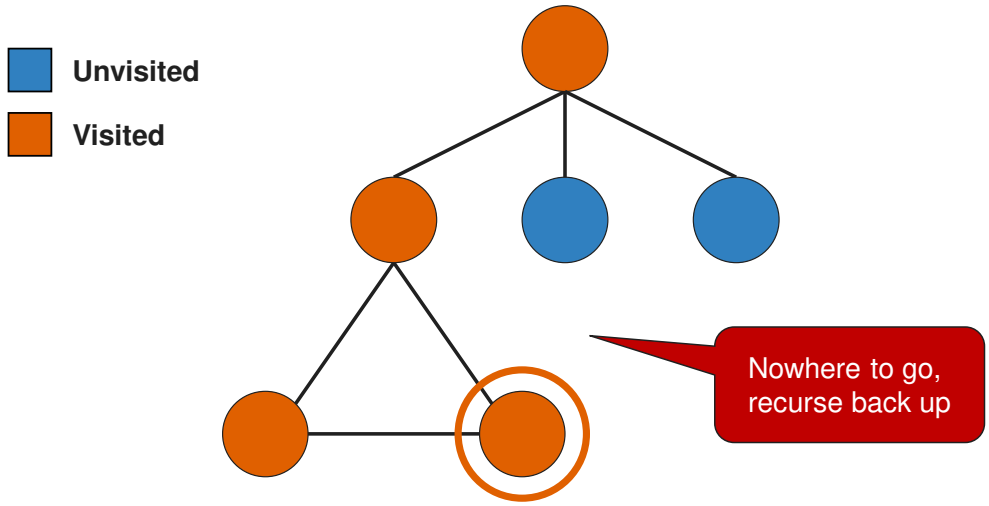
Graph Traversal



Graph Traversal



Graph Traversal



Graph Theory

Theoretical Framework

Theoretical Framework

- *Graph Theory* is based directly on set theory
- So, all the material you learned in Discrete Mathematics (CSC 28) is applicable to graphs

Theoretical Framework

- This section will introduce some of the theories.
- You will cover these extensively in CSC 135 (the next course)

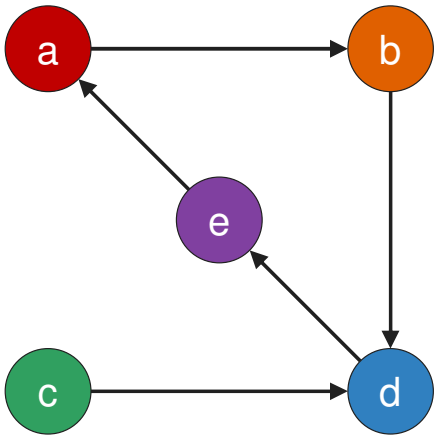
Formal Definition

A graph $G = (V, E)$ is defined by an order-pair which consists of two sets:

1. <u>finite</u> set V of items called *vertices*
2. <u>finite</u> set E of ordered-pairs called *edges* which contains items in V

Set of vertices V

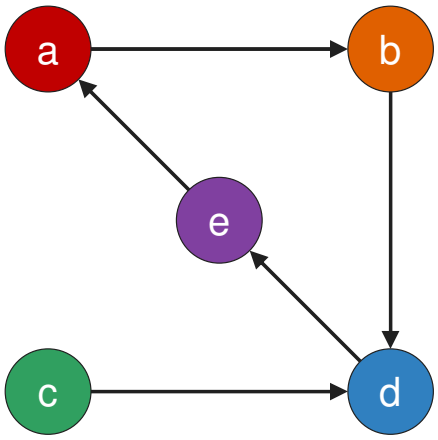
- a
- b
- c
- d
- e



Set of edges E

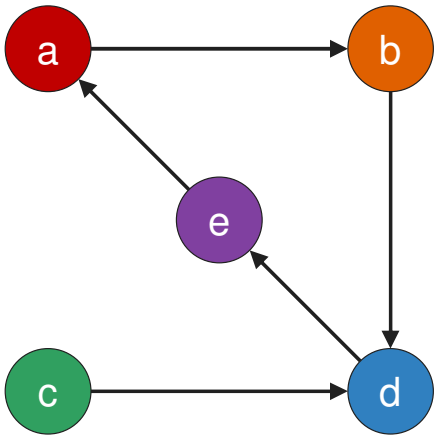
- (a, b)
- (b, d)
- (c, d)
- (d, e)
- (e, a)

Notice that the ordered-pairs have the source and destination



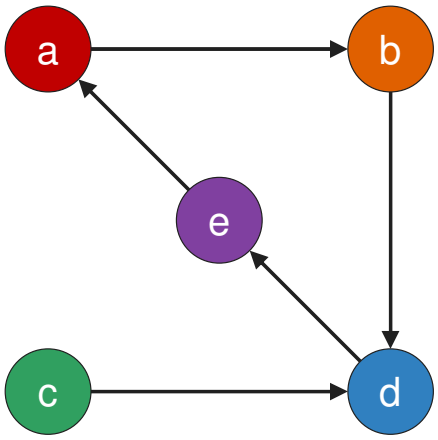
Edges are Tuples

- Edges are represented as ordered-pairs (*2-tuple*)
- Unlike a set, the order of items is meaningful
- So, there is a specific source and destination



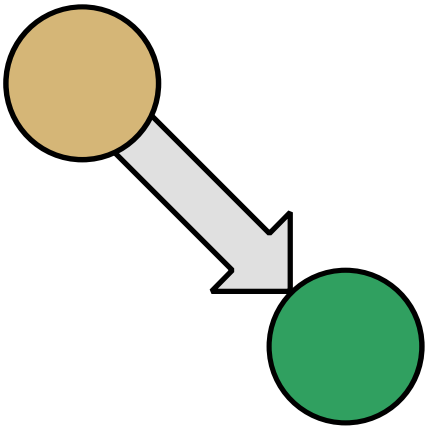
Edges are Tuples

- A edge set E contains ordered pairs
- Each of the two items are of the same type, which is a *relation-on*
- As such it creates a *relationship*



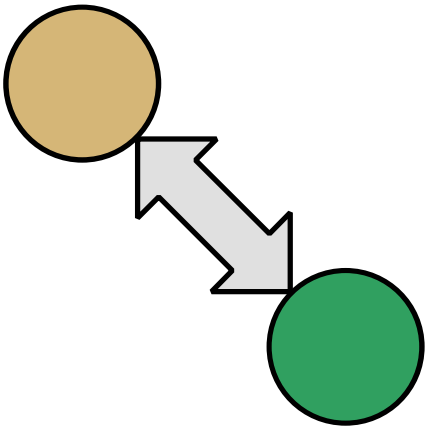
Directed Graphs

- So, the definition of graphs naturally specifies a directed graph
- Each tuple edge has a specific source and destination



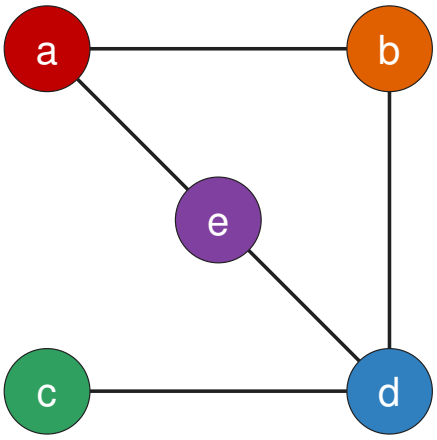
Undirected Graphs

- Undirected graphs have edges that have the <u>same</u> meaning for both directions
- If there is an edge (a, b) then there is also an edge (b, a)
- In other words, the edge relationship is *symmetric*



Undirected Graph: Symmetric Edges

- (a, b) (b, a)
- (b, d) (d, b)
- (c, d) (d, c)
- (d, e) (e, d)
- (e, a) (a, e)

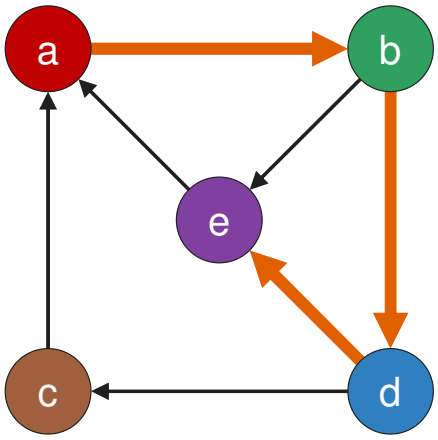


Paths

- A path is also based on the relationship of the edges
- If the edge set is *transitive* then it is a path

Example Path Edges

(a, b) → (b, d) → (d, e)

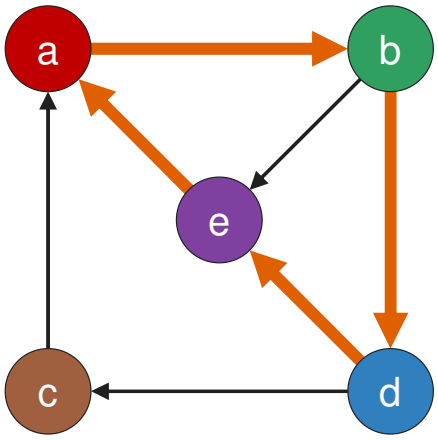


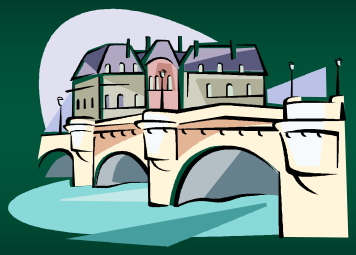
Paths

- A cycle is a slight variation of the path
- Like before, it is a *transitive* relationship, but the second item in the last tuple is the first item in the initial tuple

Example Cycle Edges

(a, b) → (b, d) → (d, e) → (e, a)





Birth of Graph Theory

... and, later, computer science!

Birth of Graph Theory

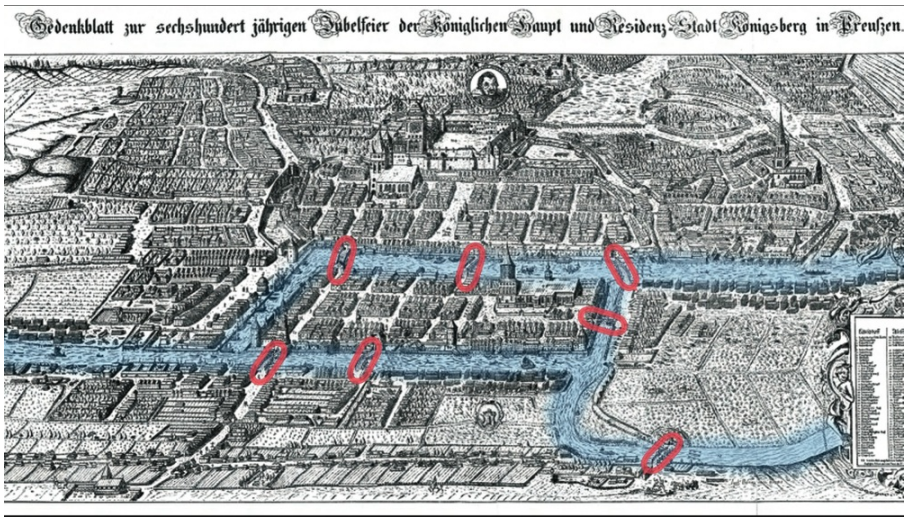
- Graph theory was created due to a perplexing question troubling mathematician *Leonhard Euler*
- Lived in Konigsberg
  - now "Kaliningrad" in Russia
  - city occupied 2 islands plus areas on both banks
  - 7 bridges over the Pregel River

Birth of Graph Theory

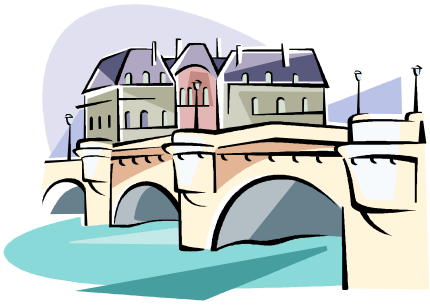
- People wondered they could:
  - leave home…
  - cross every bridge <u>once</u>
  - and return to their starting point
- This is known as the *Konigsberg Bridge Problem* & was unsolved in the 1700's

Konigsberg Bridge Problem

Konigsberg Bridge Problem

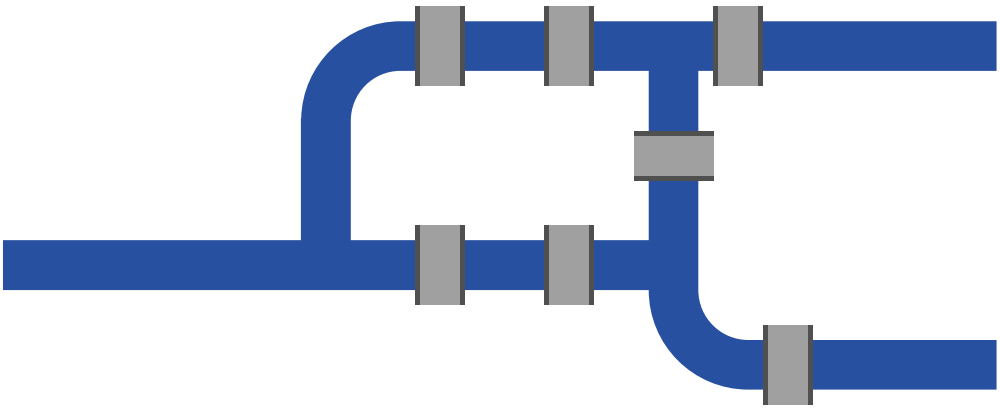


Konigsberg Bridge Problem

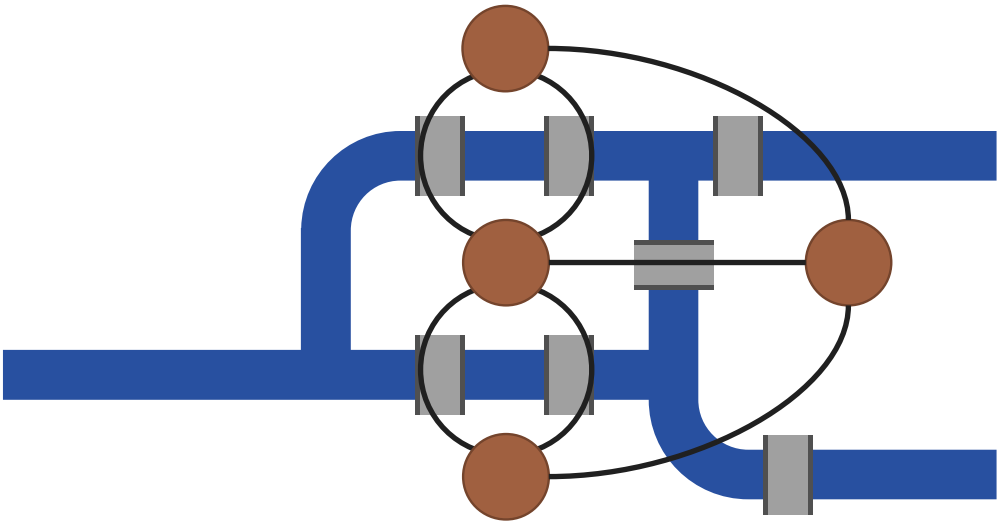


- The problem reduces to 4 points and several links to between the points
- From this, Euler created the first graph and began the study of their properties

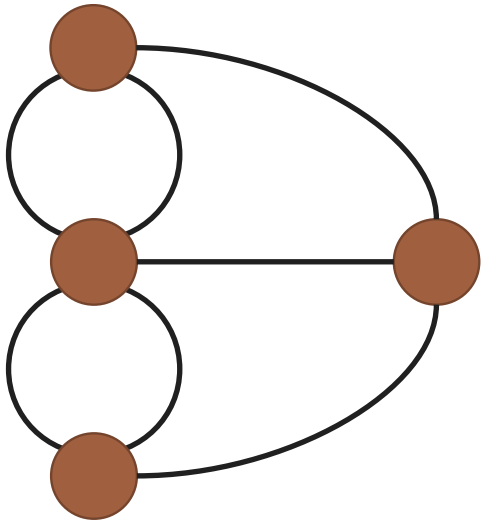
Konigsberg Bridge Problem



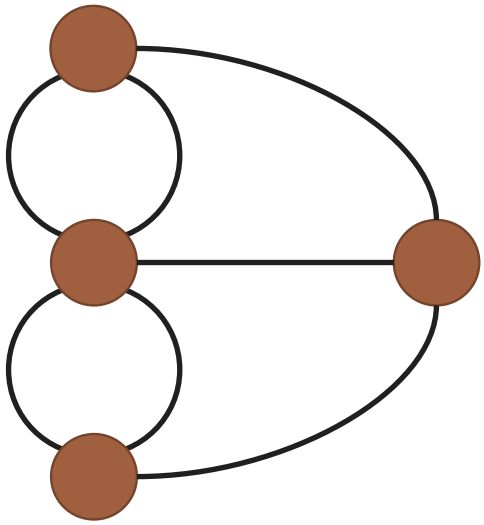
Konigsberg Bridge Problem



Konigsberg Bridge Problem

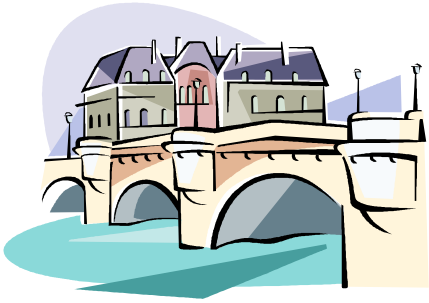


Konigsberg Bridge Problem



The Solution to Konigsberg

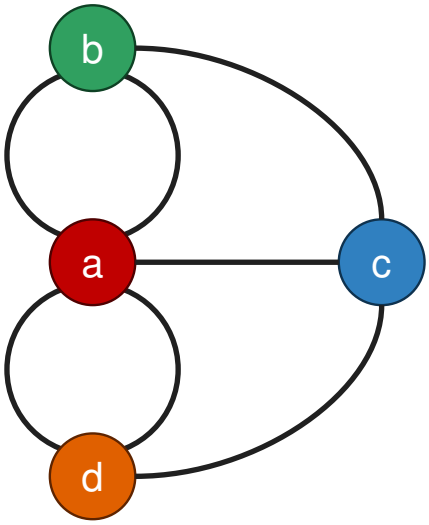
- In 1736, Euler proved that no such traversal exists
- *Eulerian circuit*, in a graph...
  - is cycle containing all the edges in the graph
  - and only traversing each edge once



The Solution to Konigsberg

- Euler proved:
  - a graph may have an Eulerian circuit if and only if there is no vertex with an odd number of edges
- Konigsberg Bridge Problem
  - 4 vertices, all with an odd number of edges
  - Sorry people of Konigsberg, there is no solution!

Konigsberg Undirected Graph

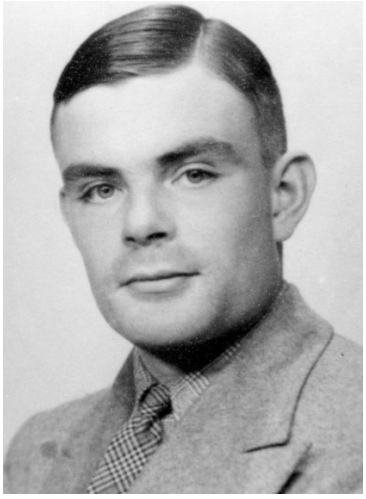



```
(  { a, b, c, d }

   { (a, b) (b, a)
     (a, b) (b, a)
     (a, d) (d, a)
     (a, d) (d, a)
     (a, c) (c, a)
     (b, c) (c, b)
     (d, c) (c, d) }  )
```

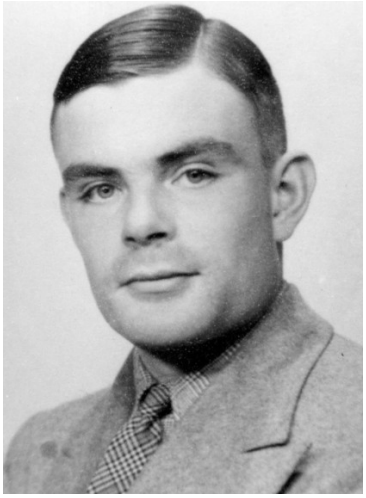

Alan Turing

- Mathematician, logician & cryptographer
- *Father of Computer Science*
  - Highest award in Computer Science is the *Turing Award*
  - Developed Turing Machines



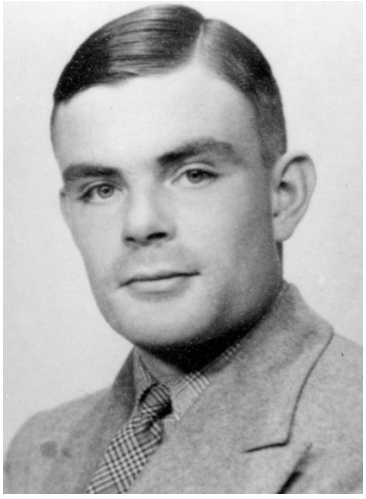
Major Work: Turing Machines

- Invented in 1937
- Logical model – not an actual computer or machine
- Based on 2 graphs (and sets on each of the edge)



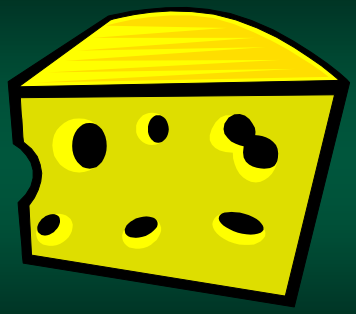
Major Work: Turing Machines

- One graph is simple array, but the other could be anything
- From this, he proved programming



Major Work: Turing Test

- Used in artificial intelligence
- Consists of a human operator *texting* a human or computer
- If the operator can't ascertain if it is a computer or human, the computer is "intelligent"
- No computer has passed it

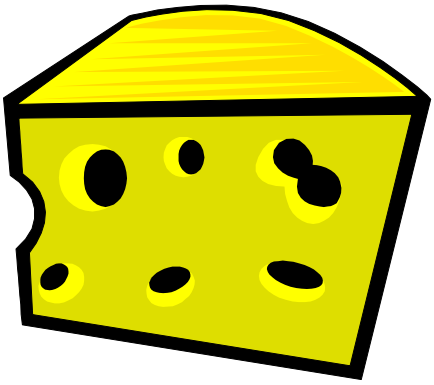


Real World Examples

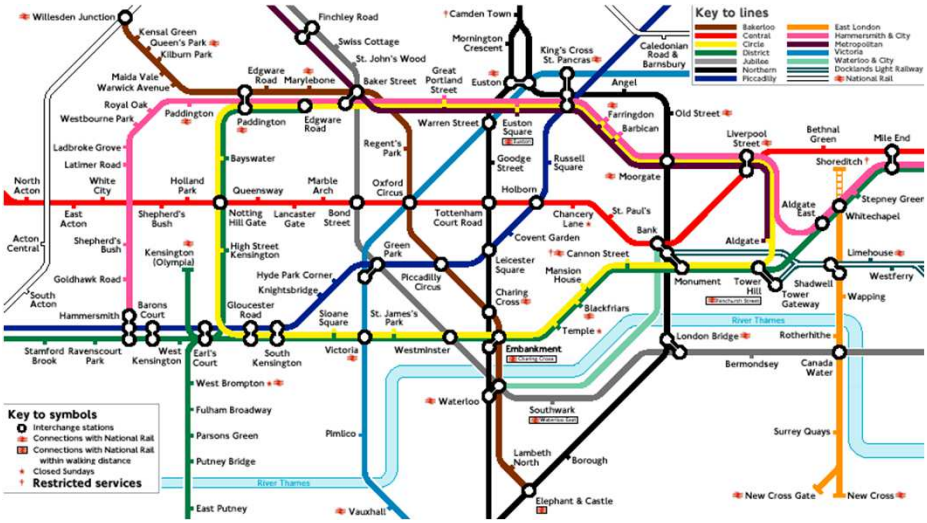
The origin and the usage

Real World Examples

- How can we lay cable at minimum cost to make every network reachable from every other?
- What is the fastest route from the national capital to each state capital?
- How can *n* jobs be filled by *n* people with maximum total utility?

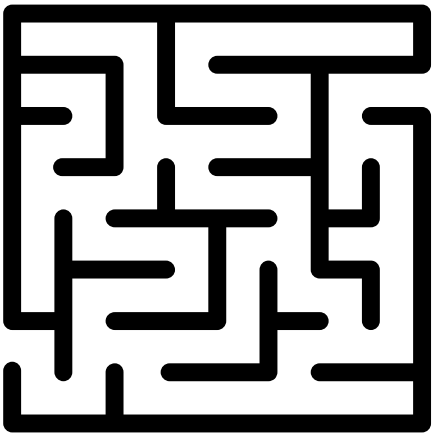


The London Underground Subway



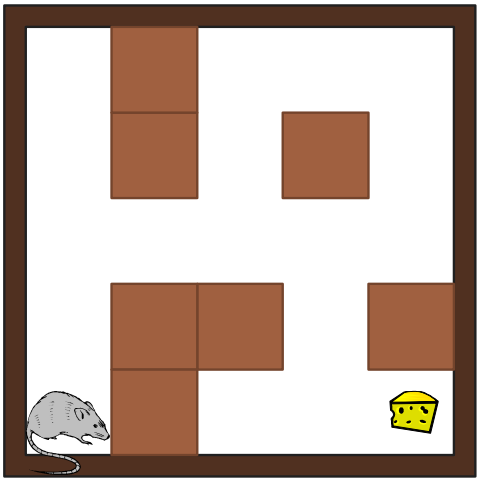
Maze Traversal

- One example of where a graph is useful is a maze traversal
- Basically, any maze can be represented with a graph
- ... and this is not so much different to how networks actually work
- ... a source must find a destination through various vertices



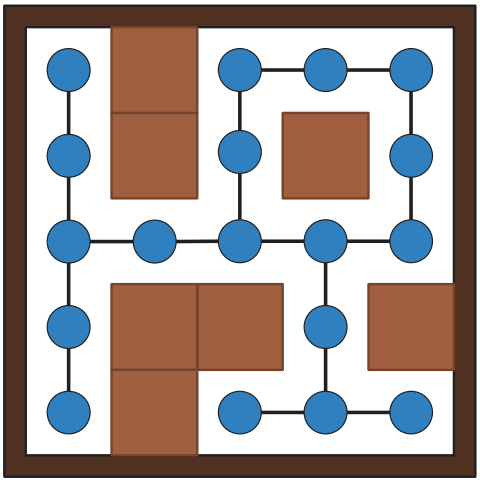
Maze Traversal

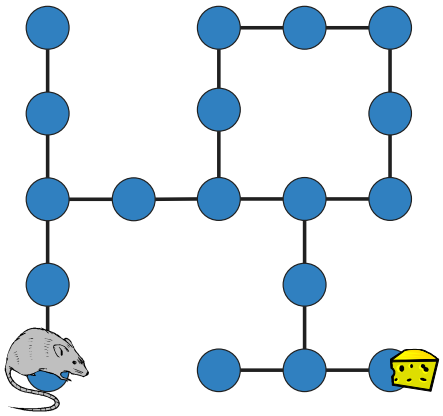
- This is a simple maze – though not to the mouse!
- We can help him find the cheese if we convert this to a graph



Maze Traversal

- The empty spaces are vertices
- The bordering ones are connected with an edge
- Is this a directed graph?



Maze Traversal

- So, to help the mouse, we can get to depth-first search on the maze
- If we find it, we can print off the vertices post-order